

Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every 7 Months

Thomas Kwa, Ben West, Joel Becker, et al.
Model Evaluation & Threat Research (METR)

A New Metric: 50% Task-Completion Time Horizon

Provides an intuitive, human-grounded measure of real-world AI capability that allows comparison across vastly different

Step 1: Task Suite



Assemble a diverse suite of
170 software/research tasks.

Step 2: Measurement



Time skilled humans and
measure AI agent success rates.

Step 3: Modeling



Fit logistic model to find task
duration where AI succeeds 50%.

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: “Which file is a shell script?” Choices: “run.sh”, “run.txt”, “run.py”, “run.md”
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

Source: METR Methodology Overview

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

A comprehensive baseline based on over 800 human trials totaling 2,529 hours of execution.

HCAST

97 diverse software tasks

Range: 1 min – 30 hrs

RE-Bench

7 difficult ML research tasks

Range: All 8 hrs

SWAA

66 short single-step tasks

Range: 1 sec – 30 sec

Family / Task	Human Time	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?"
wikipedia_research	1 minute	Research simple factual information from Wikipedia
oxdna_simple	9 minutes	Detect and fix a bug in molecular dynamics simulation input files
munge_data	56 minutes	Write a Python script to transform JSON data by inferring rules
cuda_backtesting	8 hours	Implement custom CUDA kernels for 30x speedup of a backtesting tool

AI Task Horizon Grew from 2 Seconds (GPT-2) to ~1 Hour 50 Minutes (o3)

The central finding is a highly consistent exponential growth trend with a doubling time of approximately 7 months.



Doubling Time

~207 days

(approximately 7 months)

Model Fit (R^2): **0.97**

Key Milestones (50% Success Time)



2019: GPT-2

~2 seconds



2020: GPT-3

~9 seconds



2023: GPT-4

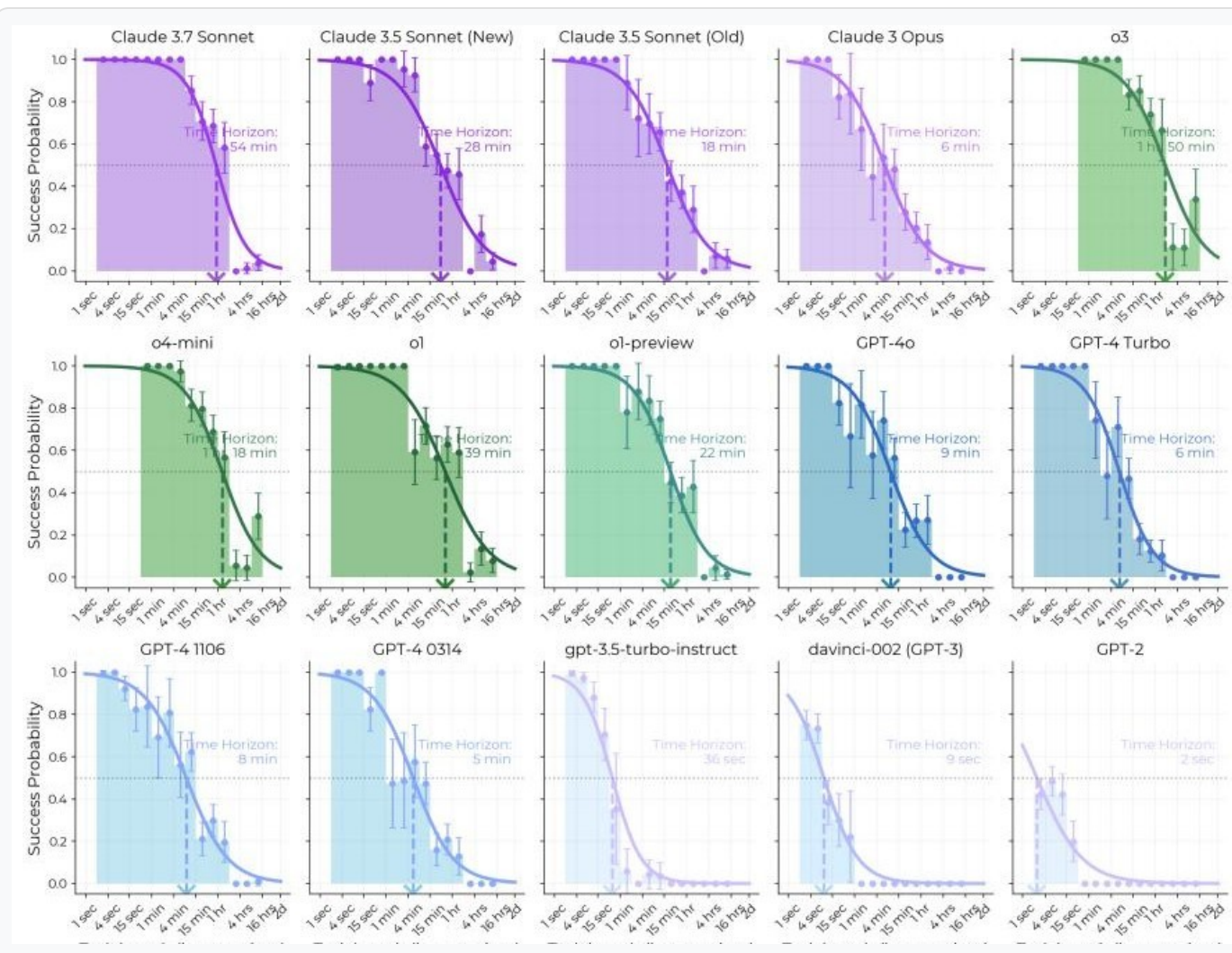
~5 minutes



2025: o3

~1 hr 50 mins

Note: 2023–2025 growth rate is ~20% faster than overall average



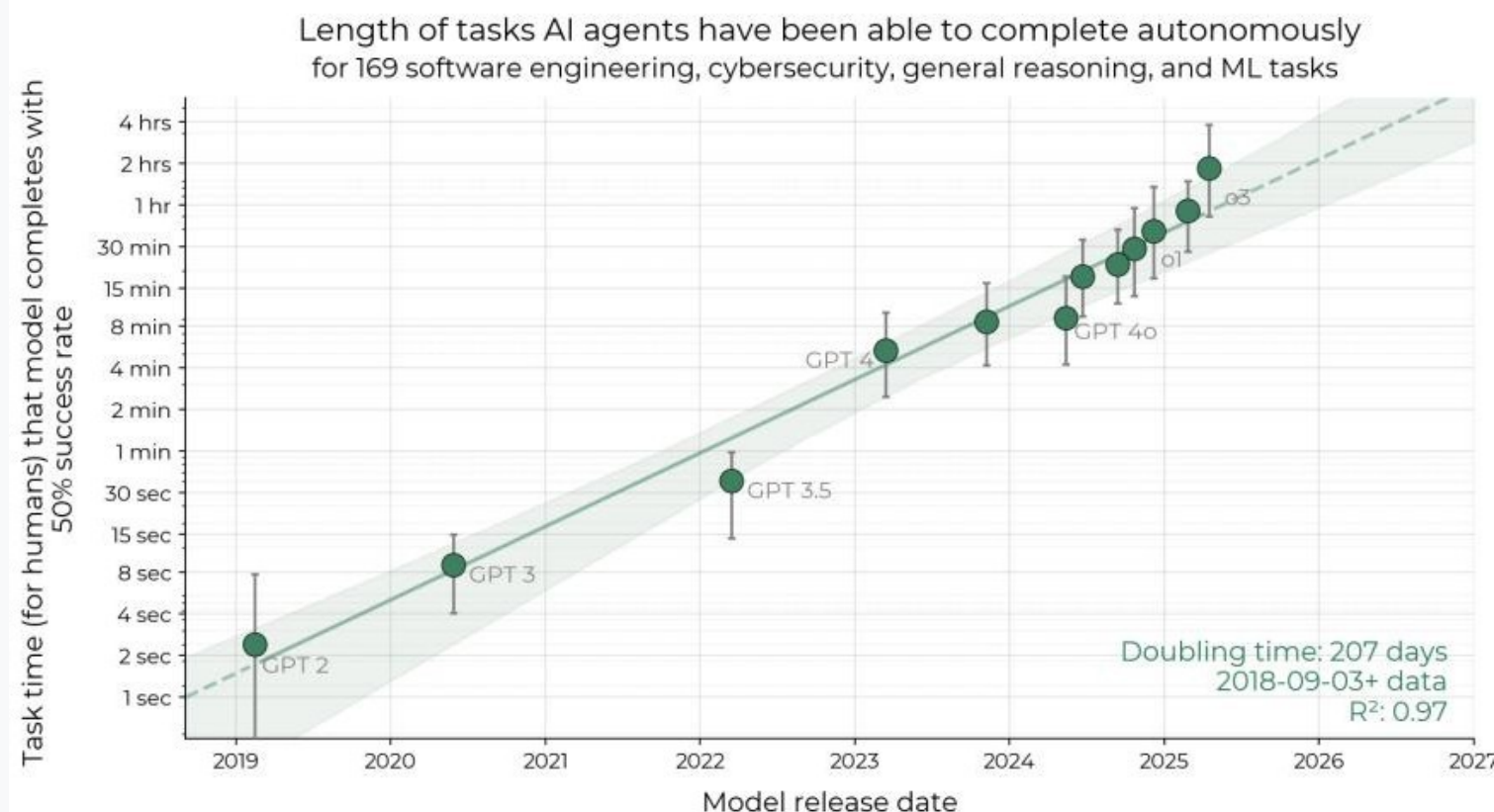
Success Probability Drops Sharply as Task Length Increases for All Models

Frontier models are pushing the curve significantly rightward, delaying the inevitable performance drop-off.

Key Observations

- **Universal Decline**
Each model's success rate declines sharply as task length and complexity increase.
- **The Frontier Shift**
Frontier models push the curve rightward compared to older models (e.g., GPT-2).

o3 (1h 50m) vs. Claude 3.7 (54m)



Source: Figure 3 (METR)

Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

While models demonstrate superior structured reasoning and tool usage, unstructured, "messier" tasks remain a critical



Core Drivers

Improved Logical Reasoning

Enhanced multi-step planning

Better Tool Use

Proficient interaction with environments



Autonomy Factors

Reliability & Self-Awareness

Consistent execution in task loops

Error Recovery

Ability to adapt to and correct mistakes



Current Limitations

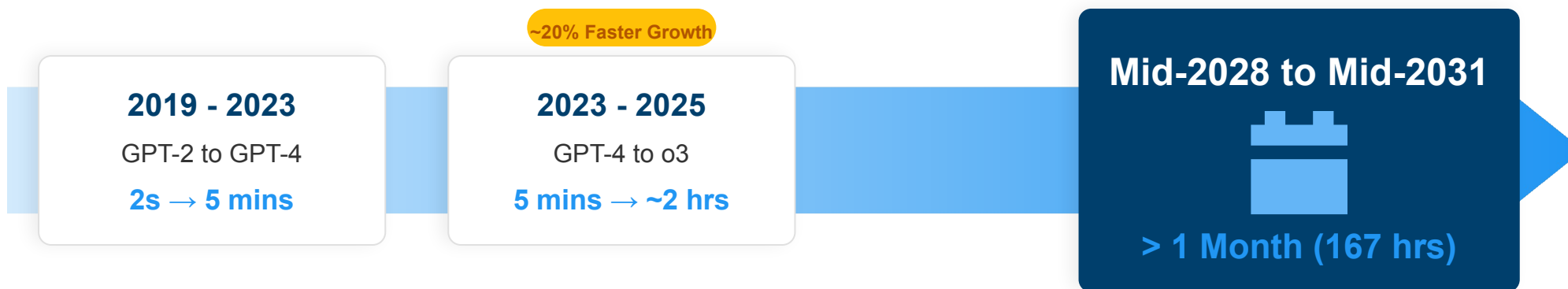
"Messier" Tasks

Performance is notably lower on less structured tasks.

External validity to perfectly represent average real-world intellectual labor is an open question.

Extrapolation Predicts Month-Long Task Automation by 2028–2031

A naive extrapolation points to models completing tasks requiring 167 human work hours (>1 month) within the next 3 to 6 years



Extrapolation Caveats

- Naive extrapolation assumes historical trends hold; architectural ceilings or data walls may slow progress.
- External validity to real-world job distributions remains an open question—automation of benchmark tasks may not perfectly map to job distributions.

Takeaways: Exponential Autonomous Capability Growth Demands Proactive S

Tracking AI autonomy via time horizon provides a robust, human-grounded signal for AI safety and workforce planning.

1 Robust Tracking Metric

The 50% time horizon is a robust, intuitive metric for tracking AI autonomy, demonstrating an $R^2 = 0.97$ over 6 years.

2 ~7-Month Doubling Time

Task completion capacity is doubling approximately every 207 days, with signs of possible acceleration (~20% faster) since 2024.

3 Current Frontier

State-of-the-art models (o3) can now autonomously complete tasks equivalent to ~2 hours of skilled human effort.

4 The Validity Question

External validity remains open.
Benchmark tasks may not perfectly represent the distribution of real-world intellectual labor.

5 Key Limitations

Current limitations include notably poor performance on unstructured tasks and uncertain generalization to chaotic environments.